MIL-G-52732/6A
27 September 1985
SUPERSEDING
MIL-G-52732/6
23 June 1972

MILITARY SPECIFICATION SHEET

GENERATOR SET, GASOLINE ENGINE DRIVEN 3 KW, 60 HERTZ,

TYPE I (TACTICAL), CLASS 2 (UTILITY)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the generator set described herein shall consist of this document and the issue in effect of specification MIL-G-52732.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for mode III, size 3 sets, model MEP-016A.

1.2 Classification.

1.2.1 Set rating. The mode III set shall be rated 0.8 power factor (PF) lagging and shall be connectable for the following voltage ratings:

a. 1 phase, 2 wire, 120 volts.
b. 1 phase, 2 wire, 240 volts.
c. 3 phase, 3 wire, 120 volts line-to-line.
d. 3 phase, 4 wire, 120 volts line-to-neutral and 208 volts line-to-line (120/208 volts).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research and Development Center, ATTN: STRBE-DS, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6115

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:31Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

MILITARY

MIL-H-52112	- Heating Torches, Equipment Winterization, Instant Lighting.
MIL-G-52732	- Generator Sets, Gasoline Engine Driven, 0.5 KW thru 10 KW, 60 Hertz, 400 Hertz and 28 Volt Direct Current, Type I (Tactical), Class 2 (Utility) General Specification for.

STANDARDS

MILITARY

MS39299	- Engine Gasoline, Air-cooled, Industrial-Type, 4 Cycle, 6 Net Continuous HP.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

ME

TA13208E5800	- Generator Set, Gasoline Engine Driven, 3 KW, 60 Hz, AC, Air-Cooled, Portable, Tubular Frame, Skid Mounted.
TA13213E4430	- Winterization Kit, 3 KW, Generator Sets.

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specifications sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Requirements shall be in accordance with MIL-G-52732, and as specified herein.

3.2 Description. The set shall be in accordance with Top Assembly TA1320E5800 and as specified herein.

3.3 Starting and operating. Servicing, adjustment and replacement shall be as specified in table I.

Source: https://assist.dla.mil -- Downloaded: 202610-22T21:31Z
Check the source to verify that this is the currentversion before use.

3.3.1 With winterization. At ambient temperatures from -25° F (-31.7° C) to -65° F (-53.9° C), the set shall start within 15 minutes with the application of external heat from a torch conforming to MIL-H-52112, type I. The set shall meet these requirements utilizing a winterization kit in accordance with TA13213E4430.

3.4 Maximum power. The maximum power (corrected) in kilowatts of the set shall be not less than 4.5 kW when operated at rated speed, voltage, and at unity power factory.

3.5 Engine.

3.5.1 Engine design. The Government furnished engine will conform to MS39299.

3.5.2 Engine temperature. Engine temperature shall not exceed the following:

Location	Maximum Temperatures	
	First 10 hours of engine operation	After 10 hours of engine operation
Sump oil	300° F (148.9°C)	265° F (129.2° C)
Cylinder head	500° F (260° C)	475° F (246.1° C)
Exhaust	1200° F (648.9° C)	1200° F (6489° C)

TABLE I. Schedule of servicing and adjusting.

All items may be serviced before test

Item No.	Hours between service periods	Item
1	8	Any minor maintenance such as retightening and retorquing nuts, bolts and screws that can be performed without disturbing or removing adjacent components, accessories or shrouding (except top shroud, pump, intake manifold, and exhaust system).
2	Not allowed	Any major maintenance that requires removal of other components, accessories or shrouding such as oil pans, cylinder heads and connecting rods.
3	As required	Cleaning and regapping or replacing spark plugs.
4	As required	Servicing fuel filter.

Source: https://assist.dla.mil -- Downloaded: 202610-22T21:31Z
Check the source to verify that this is the current version before use.

3.5.3 Governing system.

3.5.3.1 Short term, steady-state stability (30 seconds), mode II and III sets. The short term steady-state stability bandwidth percentage shall not exceed 1 percent when using the Texas Instrument Recorder and 1.4 percent when using the Gould Recorder.

3.5.3.2 Long term steady-state stability (4 hours), mode II and III sets. The long term steady-state stability bandwidth percentage shall not exceed 2 percent when using the Texas Instrument Recorder and 2.7 percent when using the Gould Recorder.

3.6 Generator.

3.6.1 Voltage unbalance, mode II and III sets. The variation between line-to-line voltages shall not exceed 5 percent.

3.6.2 Phase balance (voltage), mode II and III sets. The difference in line-to-line and line-to-netural voltage shall not exceed 1 percent.

3.6.3 Voltage waveform, mode II and III sets. The deviation factor of the line-to-neutral and the line-to-line voltage for 3-phase generator connections at rated load, no-load and unity power factor rated kW shall not exceed 5.0 percent. Under the same conditions, the deviation factor at the single phase connections shall not exceed 6.0 percent. Single frequency harmonics shall not exceed 3.0 percent.

3.6.4 Output voltage ripple, mode IV sets. Not applicable for the mode III, size 3 set.

3.6.5 Saturation curves. With the generator temperature stabalized at normal ambient conditions the exciter field current for rated voltage shall not exceed 0.38 ampere at no-load and 0.90 ampere at rated load.

3.6.6 Voltage adjustment range. The minimum acceptable voltage adjustment range shall be 5 percent above and below rate voltage.

3.6.7 Voltage regulation. The voltage regulation percentage shall not exceed 4 percent.

3.6.8 Voltage steady-state performance. The voltage steady-state stability bandwidth percentage shall not exceed 2 percent.

3.6.9 Voltage transient performance. The voltage dip or rise percentage shall not exceed 30 percent.

3.6.10 Generator efficiency. The generator efficiency shall not be less than 80 percent at 1.0 PF and 70 percent at 0.8 PF.

3.7 Circuit interrupter. The circuit interrupter shall open within 0.25 seconds under the condition of a short circuit in any one phase or in all phases simultaneously.

Source: https://assist.dla.mil -- Downloaded: 202610-22T21:31Z
Check the source to verify that this is the currentversion before use.

3.8 Instruments. The inaccuracy or error of the metering system shall not exceed the following:

Instrument	Allowable Value
Ammeter (percent load)	10 percent inaccuracy
Voltmeter	3 percent inaccuracy
Frequency meter	1.25 Hz error

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance provisions. The quality assurance provisions shall be in accordance with MIL-G-52732 and as specified herein.

4.2 Engine preventative maintenance services. The engine preventative maintenance services shall be preformed as specified in table I.

4.3 Drop test (free fall). The height of the drop shall be 18 inches for the mode III, size 3 set.

4.4 Vibration test. The handrail or shock mount resonances shall not be utilized for resonant dwell conditions.

5. PACKAGING

5.1 Preparation for delivery. Preparation for delivery shall be in accordance with MIL-G-52732.

6. NOTES

6.1 Notes. The notes specified in MIL-G-52732 are applicable to this specification herein.

6.2 Stable operation (voltage). Stable operation is operation within a bandwidth of 2 percent of the rated voltage.

6.3 International standardization. Not applicable for the mode III, size 3 set.

Custodians:
Army - ME
Air Force - 99
Navy - YD

Review activities:
Army - ER, GL
Air Force - 80
DLA - GS

Preparing activity:
Army - ME

Project 6115-0238

Source: https://assist.dla.mil -- Downloaded: 202610-22T21:31Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for submitting comments and suggestions for improvements. All users of military standardization documents are invited to suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE* mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretati too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate th problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, a acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are bein considered.

NOTE. This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification o specification requirements on current contracts. Comments submitted on this form do not constitute or imply authoriza to waive any portion of the referenced document(s) or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE ARMY

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12062 WASHINGTON D.C.
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

USA Belvoir Research and Development Center
Attn: STRBE-DS
Fort Belvoir, VA 22060-5606

NO PO
NECES
IF MA
IN T
UNITED

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:31Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 202610-22T21:31Z
Check the source to verify that this is the current version before use.

posing and shall be responsible for the following voltage tapings: (5)